Introduction to Intensive Programming in C++

Lecture 4: Greedy algorithms, codes, and trie

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Contents

Greedy algorithms

General strategy
It is a technique, not an algorithm.

Pick the most obvious solution!

proof strategy !!!

- Assume that there is an optimal solution that is different from the greedy solution.
- Find the *first* difference between the two solutions.
- *Exhange argument*: Argue that we can exchange the optimal choice for the greedy choice without making the solution worse (although the exchange might not make it better).

Greedy algorithms

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It is a technique, not an algorithm.

Pick the most obvious solution!

proof strategy !!! ← this is the hard part

- Assume that there is an optimal solution that is different from the greedy solution.
- Find the *first* difference between the two solutions.
- *Exhange argument*: Argue that we can exchange the optimal choice for the greedy choice without making the solution worse (although the exchange might not make it better).

- Greedy cannot handle problems with (many) local mins
- Exchange argument (essentially) proves that there is no local min

Other greedy algorithms

- Coin exchange
- Optimal caching
- *Minimum spanning tree (Kruskal, Prim)*
- *(single pair) Shortest path*

Optimal chaching

Problem: n files of length $L[1..n]$ are stored sequentially on a tape T

cost of accessing the k-th file is $\text{cost}(k) = \sum_{i=1}^{k} L[i]$

Store the files in a way to minimize the expected cost

Optimal chaching

Problem: n files of length $L[1..n]$ are stored sequentially on a tape T

cost of accessing the k-th file is $\text{cost}(k) = \sum_{i=1}^{k} L[i]$

Store the files in a way to minimize the expected cost

If every file is equally likely, then $\mathbb{E}[\text{cost}] = \sum_{k=1}^{n} \frac{\text{cost}(k)}{n} = \frac{1}{n} \sum_{k=1}^{n} \sum_{i=1}^{k} L[i]$

BUT different orders give different expected costs

Consider a permutation π then

$$\mathbb{E}[\text{cost}(\pi)] = \frac{1}{n} \sum_{k=1}^{n} \sum_{i=1}^{k} L[\pi(i)]$$

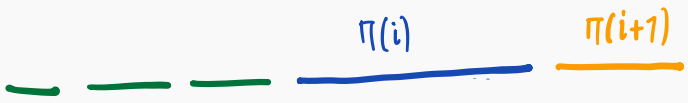
SORT the files in increasing length

GREEDY: put the cheapest to access file first

WHY ?

Optimal chaching

LEMMA: $\min \mathbb{E}[\text{cost}(\pi)]$ if $L[\pi(i)] \leq L[\pi(i+1)] \quad \forall\, i$

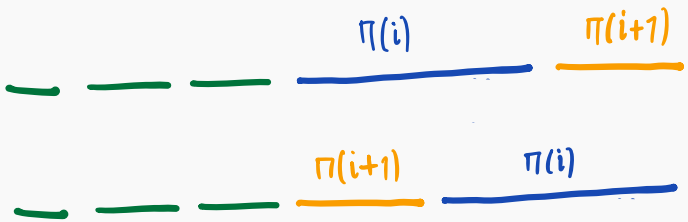


Optimal chaching

LEMMA: $\min \mathbb{E}[\text{cost}(\pi)]$ if $L[\pi(i)] \leq L[\pi(i+1)] \quad \forall\, i$

proof:

Assume $\exists\, i$ s.t. $L[\pi(i)] > L[\pi(i+1)]$



then if we swap $\pi(i)$ and $\pi(i+1)$

- the cost of $\pi(i)$ increases (+) by $L[\pi(i+1)]$
- the cost of $\pi(i+1)$ decreases (-) by $L[\pi(i)]$

the difference is $L[\pi(i+1)] - L[\pi(i)] \leq 0$

so we reduce the cost, if we swap pairs that are not in order.

Optimal chaching

Variant : n files , lengths $L[1..n]$, frequences $F[1..n]$

$$Fcost(\pi) = \sum_{k=1}^{n} \left(F[\pi(k)] \cdot cost(\pi(k)) \right) = \sum_{k=1}^{n} \left(F[\pi(k)] \cdot \sum_{i=1}^{k} L[\pi(i)] \right)$$

GREEDY : But how to sort now ?

Optimal chaching

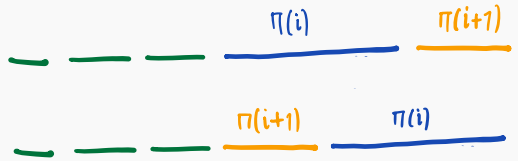
Variant : n files, lengths $L[1..n]$, frequences $F[1..n]$

$$Fcost(\pi) = \sum_{k=1}^{n} \left(F[\pi(k)] \cdot cost(\pi(k)) \right) = \sum_{k=1}^{n} \left(F[\pi(k)] \cdot \sum_{i=1}^{k} L[\pi(i)] \right)$$

GREEDY : But how to sort now ?

LEMMA : $\min Fcost(\pi)$ when $\frac{L[\pi(i)]}{F[\pi(i)]} \leq \frac{L[\pi(i+1)]}{F[\pi(i+1)]} \quad \forall i$

Assume $\exists$ i s.t. $\frac{L[\pi(i)]}{F[\pi(i)]} \geq \frac{L[\pi(i+1)]}{F[\pi(i+1)]}$ then swap them



- for $\pi(i)$ the cost increases (+) by $F[\pi(i)] \cdot L[\pi(i+1)]$
- for $\pi(i+1)$ the cost decreases (−) by $F[\pi(i+1)] \cdot L[\pi(i)]$

So in total $F[\pi(i)] \cdot L[\pi(i+1)] - F[\pi(i+1)] L[\pi(i)] \leq 0$ So we reduce the cost if we swap

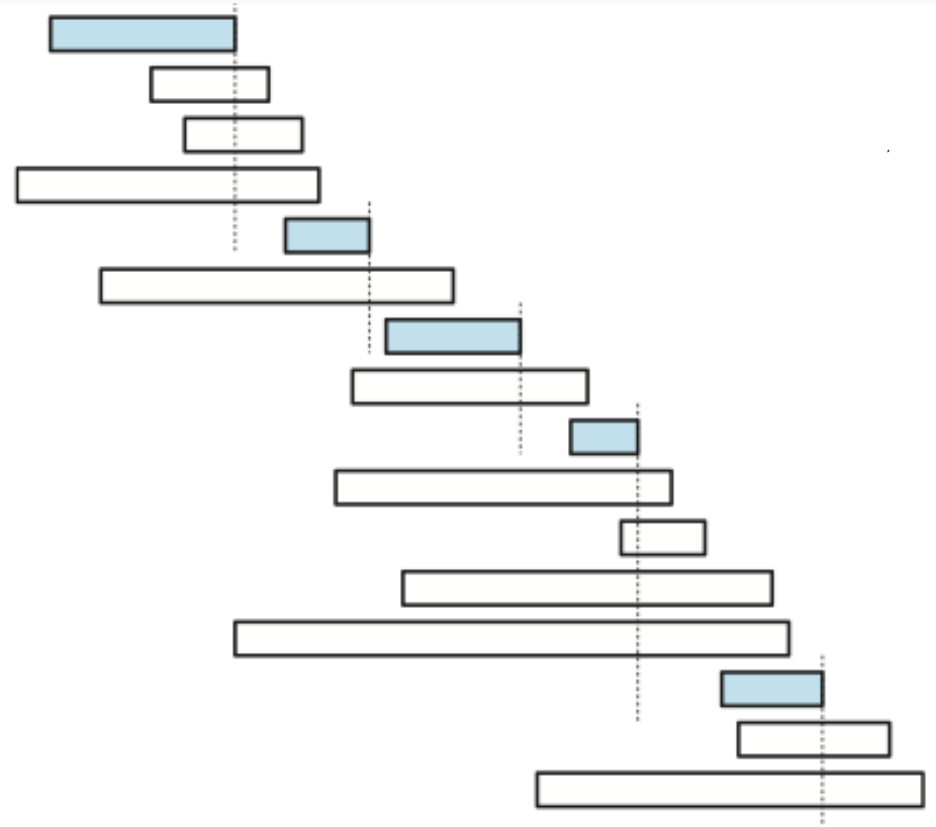
Class scheduling

n classes

starting times $S[1..n]$

finishing times $F[1..n]$

Maximize the number of classes that you attend



```
GREEDYSCHEDULE(S[1..n], F[1..n]):
  sort F and permute S to match
  count ← 1
  X[count] ← 1
  for i ← 2 to n
      if S[i] > F[X[count]]
          count ← count + 1
          X[count] ← i
  return X[1..count]
```

Class scheduling

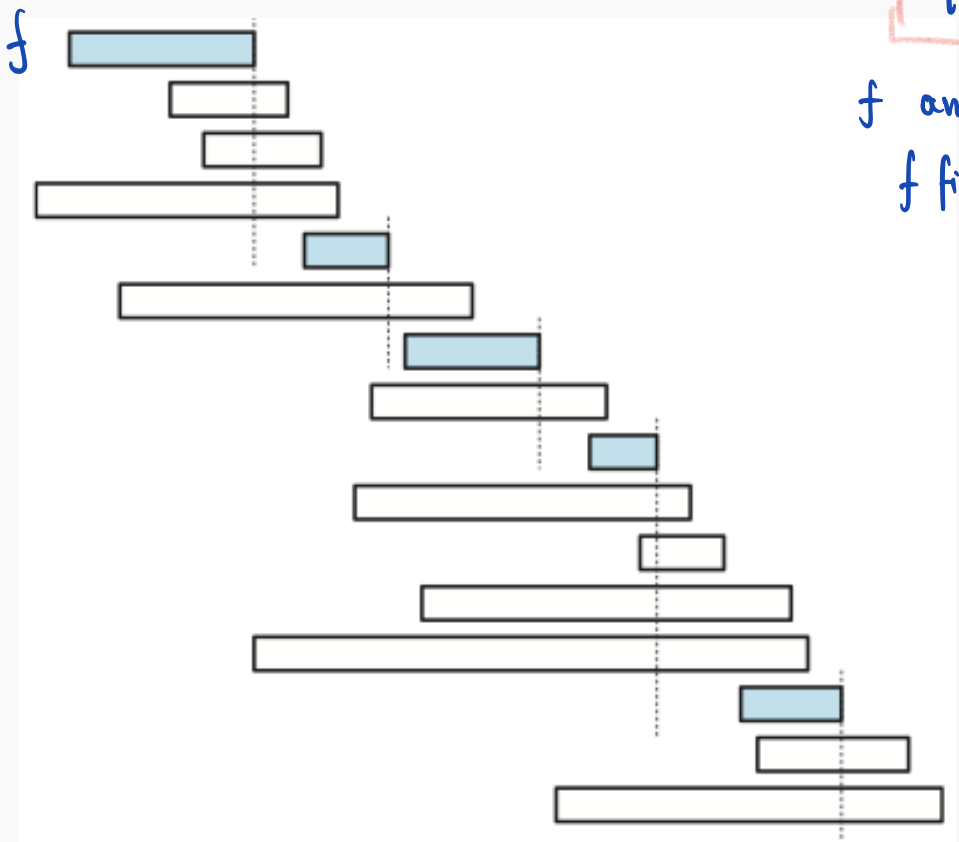
n classes

starting times $S[1..n]$

finishing times $F[1..n]$

Maximize the number of classes that you attend

At least one maximal conflict-free schedule includes the first finishing class



f and $X = \{g_1, g_2, \ldots, g_k\}$

f finishes before g_1 so $(X \setminus \{g_1\}) \cup \{f\}$ is conflict-free and optimal

```
GREEDYSCHEDULE(S[1..n], F[1..n]):
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Class scheduling

n classes

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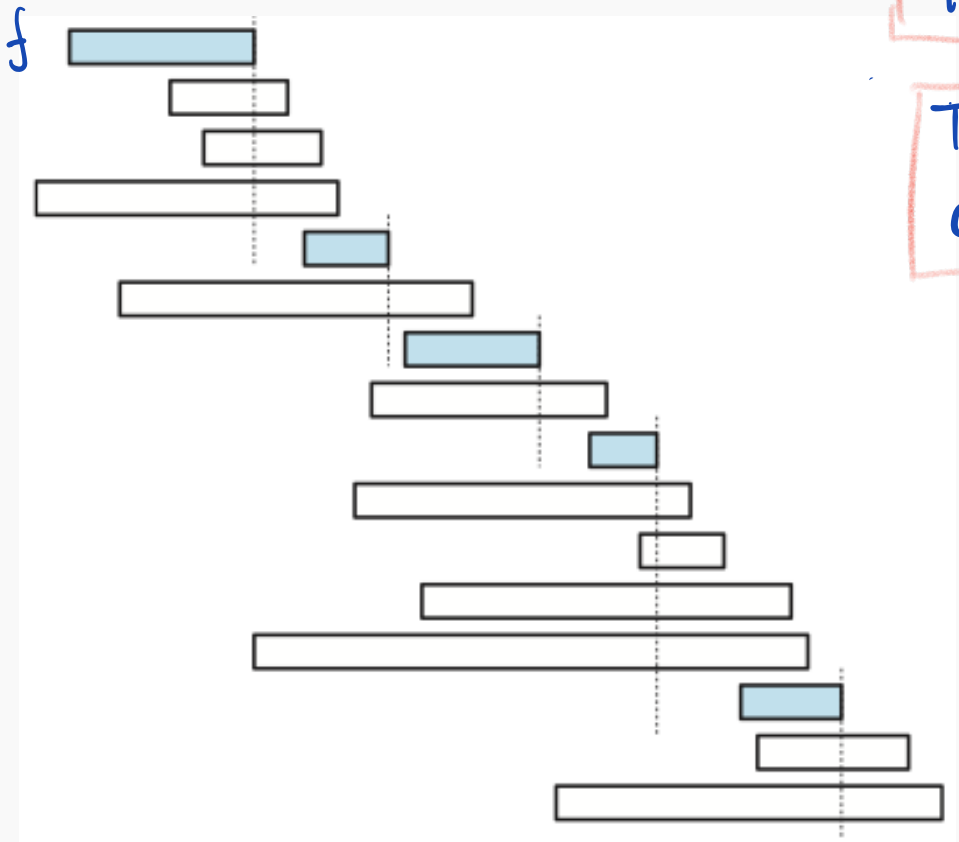
Maximize the number of classes that you attend



At least one maximal conflict-free schedule includes the first finishing class

The greedy produces an optimal conflict-free schedule

induction by generalizing

```
GREEDYSCHEDULE(S[1..n], F[1..n]):
  sort F and permute S to match
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  return X[1..count]
```

Complexity: sorting + selecting $O(n \lg n) + O(n) = O(n \lg n)$

Greedy children

There are cookies, with sizes s_j

There are children with greed factor g_i.

If $s_j \geq g_i$ we can give the cookie j to child i and the child is happy.

Maximize the number of happy children.

```
if i < s.size() and j < g.size()
   if s[i] >= g[j] {
     i++;
     j++;
     r++;
   } else {
     i++;
   }
}
```

Sort s and g (in which order?)

Minimum number of platforms

The input is the arrival and departure time of n trains in a station.

Find the mimimum number of platforms needed to accomodate all trains in the station.

- Sort the two arrays.
- Increase or decrease a counter.
- If an arrival and a departure happen at the same time, give priority to the departure.

Construct the maximum number

Given two arrays of lenght m and n representing two numbers.

Construct a maximum number with $k \leq m + n$ digits.

We should preserve the relative order of the digits of each array.

$$\mathtt{A} = [6, 7]$$

$$\mathtt{B} = [6, 0, 4]$$

$$k = 5$$

$$\mathtt{R} = [6, 7, 6, 0, 4]$$

Smallest range covering elements from k lists

You have k lists of sorted integers in non-decreasing order.

Find the smallest range that includes at least one number from each of the k lists.

$$[a, b] < [c, d] \text{ if } \begin{cases} b - a < d - c \\ \text{or} \\ b - a = d - c \text{ and } a < c \end{cases}$$

$k=3$
$n=6$

$[4, 7, 9, 12, 15]$
$[0, 8, 10, 14, 20]$
$[6, 12, 16, 30, 50]$

result $[6, 8]$

'Sliding Window'

Binary codes

constant length

char	code
A	00
B	01
C	10
D	11

enumeration

not valid

char	code
A	10
B	11
C	1011
D	111

optimal

char	code
A	0
B	110
C	10
D	111

INPUT

00 00 01 00 10 11 00 10 00

A A B A C D A C A

length of the message = 18

10 11

A B

or C

0 0 110 0 10 111 0 10 0

A A B A C D A C A

length of the message = 15

Huffman codes

Prefix-free code

No codeword is a prefix of any other codeword

Huffman codes [~ 1951]

- Let f_i be the number of occurences of letter i.
- Total number of symbols needed

$$\sum_i f_i \, \text{length}(i)$$

- Huffman algorithms finds the optimal prefix-free code.

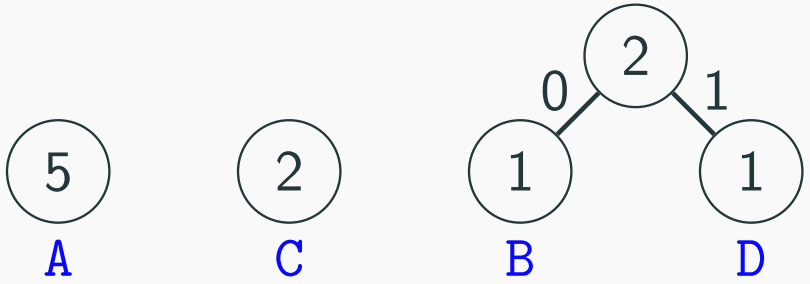
Variable length code

What does it mean?
Is there a lower bound?

extends binary trees to minimum weighted path length from given weights

Merge the two least frequent letters and recurse

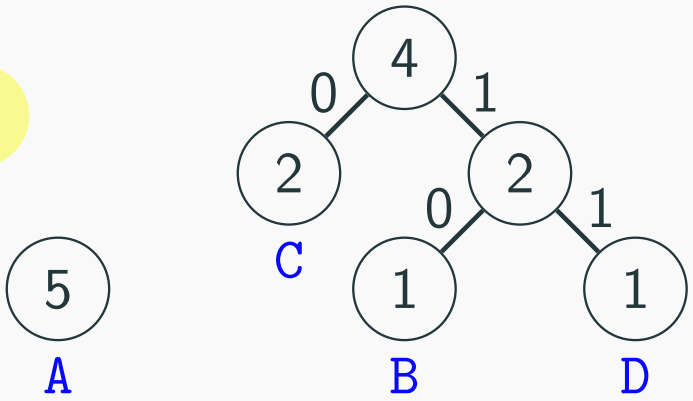
binary code and compression



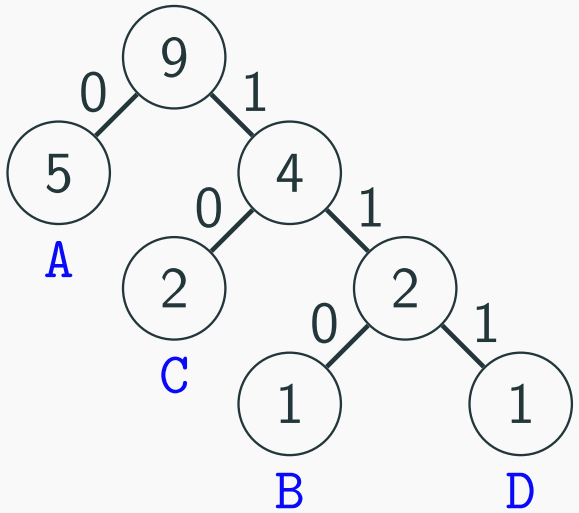
char	code
A	0
B	110
C	10
D	111

not unique

why?



AABACDACA



construction : priority queue → binary heap

Complexity : O(n) queries, each costs O(lgn)

O(nlgn)

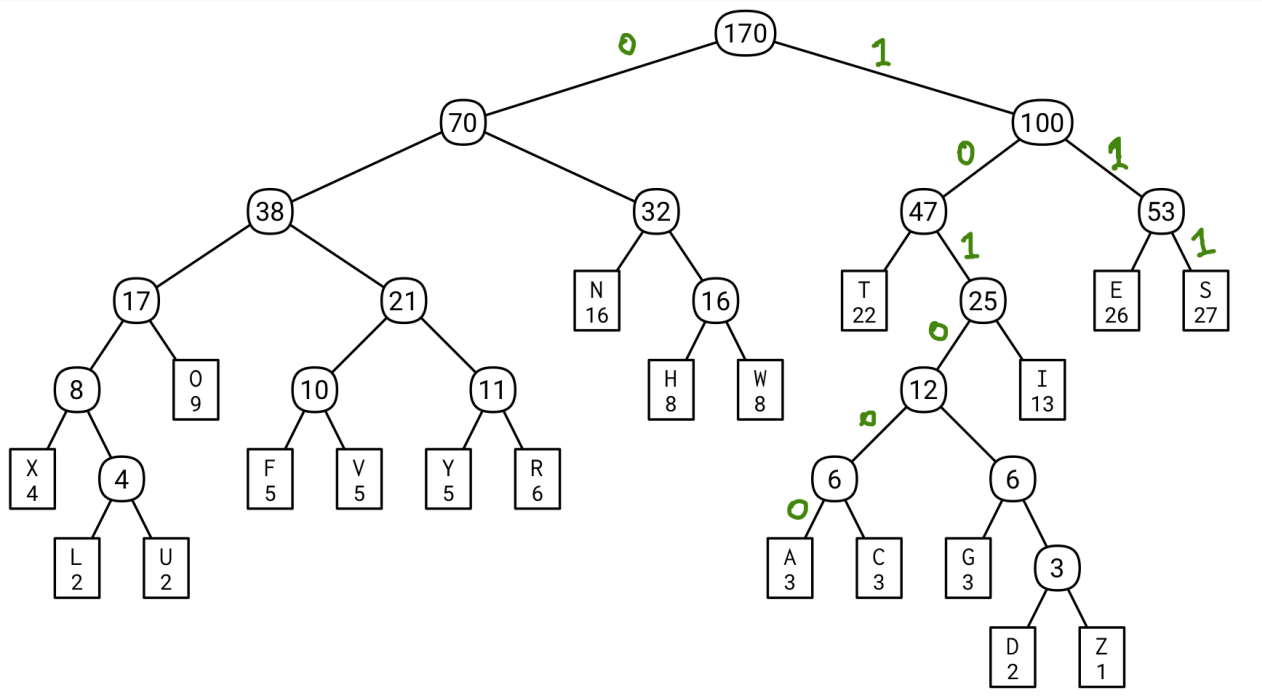
Demo Huffman tree

Hauffman codes : another example

This sentence contains three a's, three c's, two d's, twenty-six e's, five f's, three g's, eight h's, thirteen i's, two l's, sixteen n's, nine o's, six r's, twenty-seven s's, twenty-two t's, two u's, five v's, eight w's, four x's, five y's, and only one z.

```
THISSENTENCECONTAINSTHREEASTHREECSTWODSTWENTYSIXESFIVEFST
HREEGSEIGHTHSTHIRTEENISTWOLSSIXTEENNSNINEOSSIXRSTWENTYSEV
ENSSTWENTYTWOTSTWOUSFIVEVSEIGHTWSFOURXSFIVEYSANDONLYONEZ
```

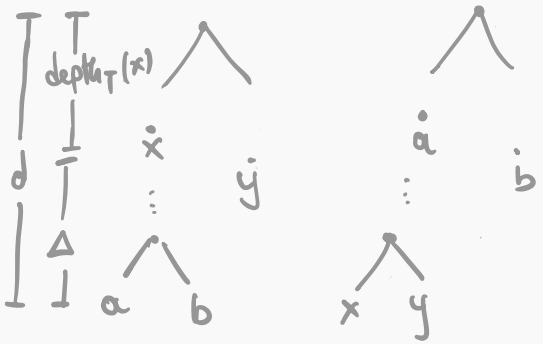
A	C	D	E	F	G	H	I	L	N	O	R	S	T	U	V	W	X	Y	Z
3	3	2	26	5	3	8	13	2	16	9	6	27	22	2	5	8	4	5	1



Huffman Codes — some details

LEMMA: Let x, y be two least frequent letters (break ties arbitrarily)
$\exists$ optimal code tree where x, y are leafs at the largest possible depth.

optimal T | new tree T'



$\Delta = d - \text{depth}_T(x)$

after swap:

- the depth of x increases by Δ (+) (cost: $+\Delta f[x]$)
- the depth of a decreases by Δ (−) (cost: $-\Delta f[a]$)

$$\text{cost}(T') = \text{cost}(T) + \Delta f[x] - \Delta \cdot f[a] = \text{cost}(T) + \Delta\underbrace{(f[x] - f[a])}_{\text{negative}}$$

But $f[x] \le f[a]$ so $\text{cost}(T') \le \text{cost}(T)$

However T optimal, thus $\text{cost}(T) \le \text{cost}(T')$

$\Big\}\ \text{cost}(T) = \text{cost}(T')$

Thm Every Huffman code is optimal prefix-free binary code

Information Theory

$A = \{a_1, \dots, a_n\}$ random variable $X \in A$

probabilities p_1 p_n $\sum p_i = 1$

Entropy $H(X) = -\sum_{x \in A} p(x) \lg p(x)$

Proposition: $0 \leq H(X) \leq \lg |A|$

for a r.v. X we need $H(X)$ bits to describe it on the average

LZW Lempel - Ziv - Welch

fixed length 12 bits encoding

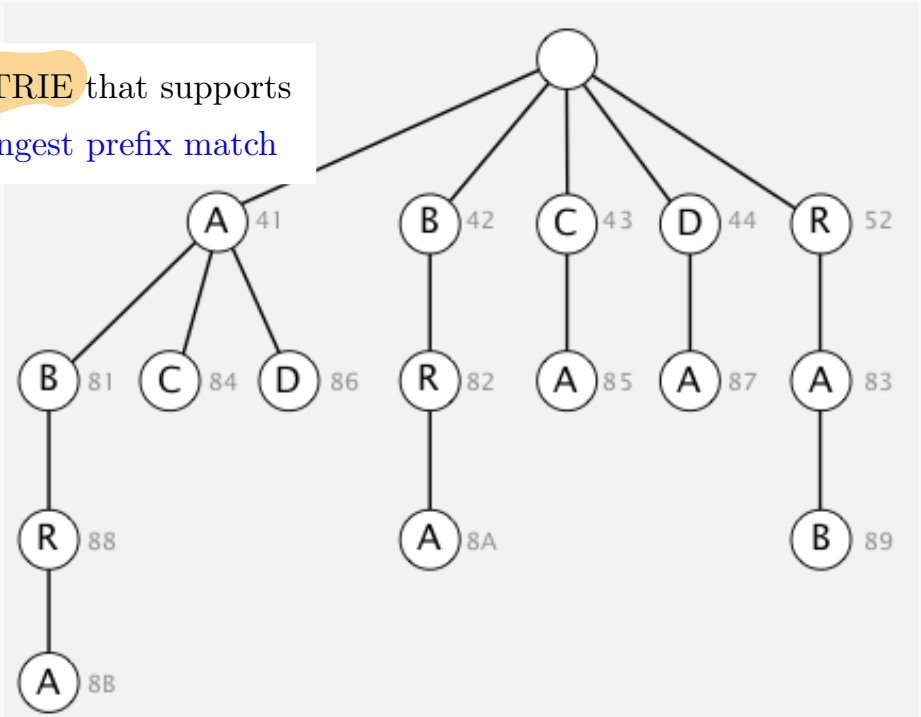
0-255
8 bits for one letter
4 bits for sequences
256-4095

input	A		B		R	A	C	A	D	A	B	R	A	B	R	A	B	R
matches	A		B		R	A	C	A	D	A B		R A		B R		A B R		
value	41		42		52	41	43	41	44	81		83		82		88		

key	value
⋮	⋮
A	41
B	42
C	43
D	44
⋮	⋮

key	value
AB	81
BR	82
RA	83
AC	84
CA	85
AD	86

a TRIE that supports longest prefix match

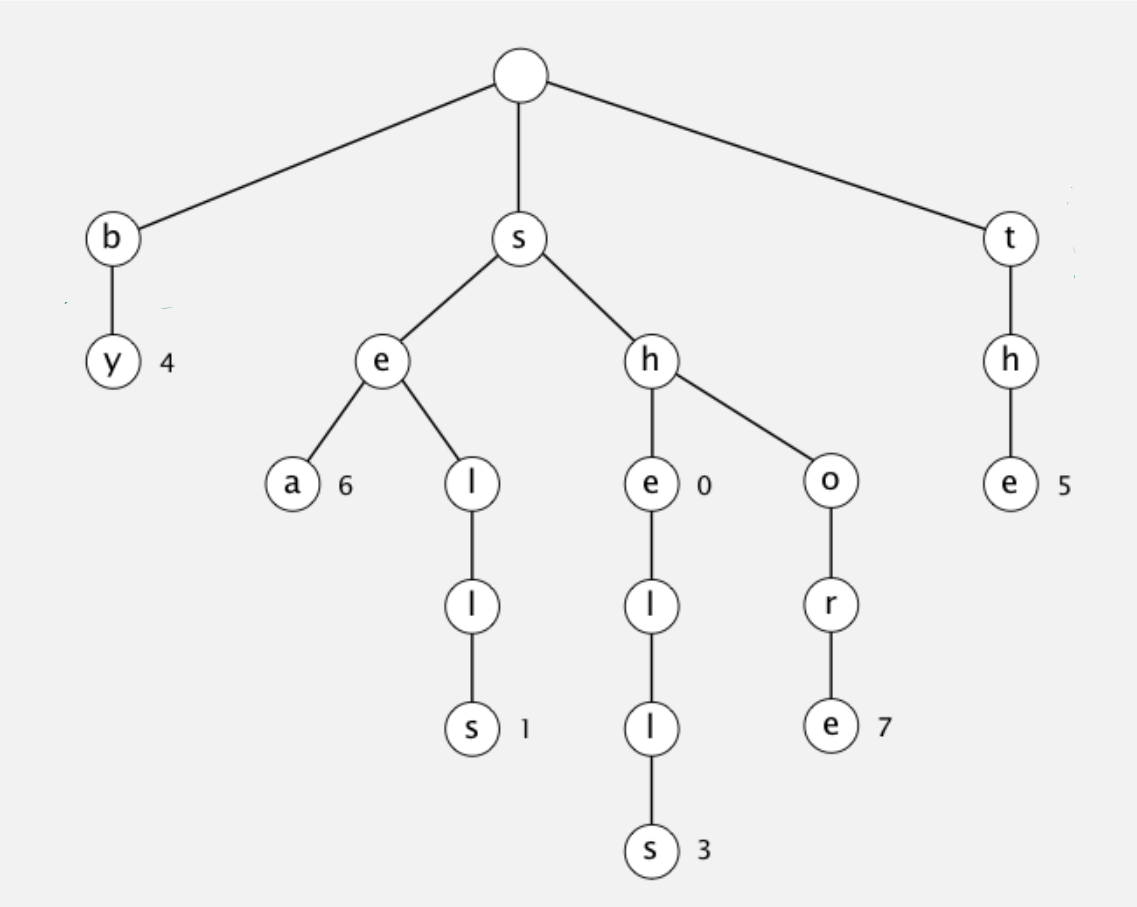


it can encode combinations

Contents

Trie: a prefix tree from reTRIEval

- You prononouce it like "tree", not like "try" (most of the times).
- Every node has (at most) r children
  (i.e., the number of symbols in the alphabet)



TQ

Can you convert any tree to a binary tree?

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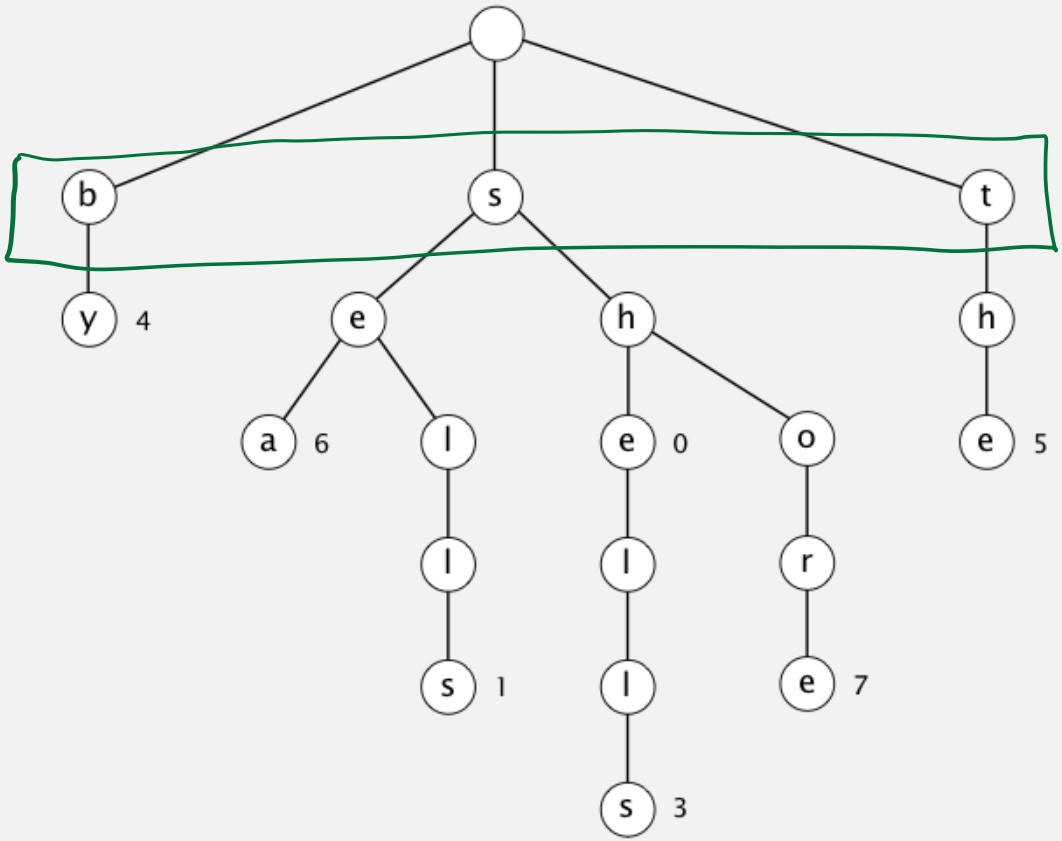
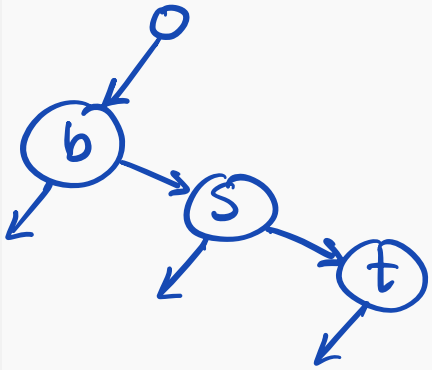


r-way tree



All the trees are binary trees

TQ

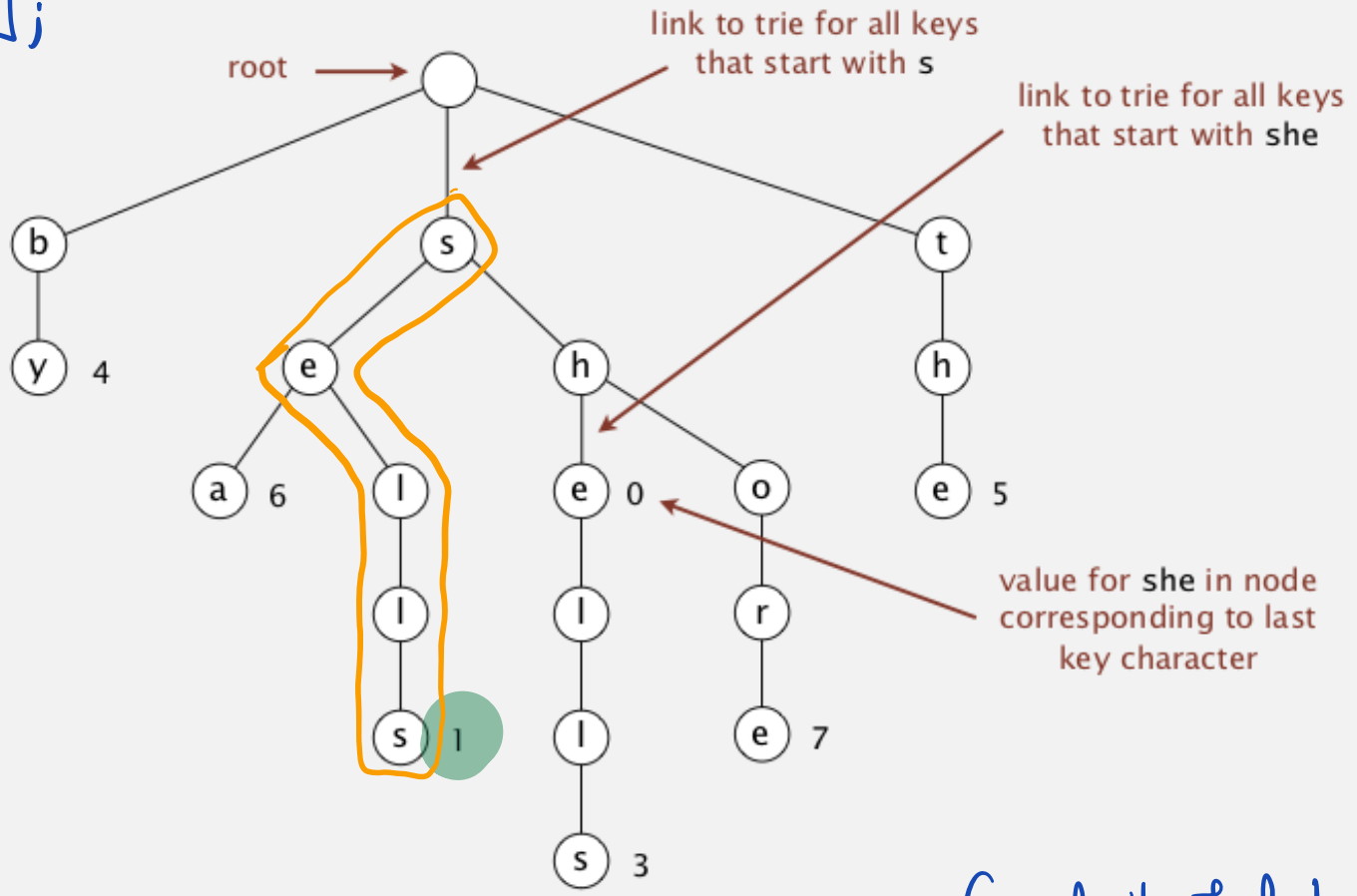
Can you convert any tree to a binary tree?

Complexity of constructing it?

Trie: an example

Implementation

```
struct Trie {
    Trie* children[r];
}
```



key	value
by	4
sea	6
sells	1
she	0
shells	3
shore	7
the	5

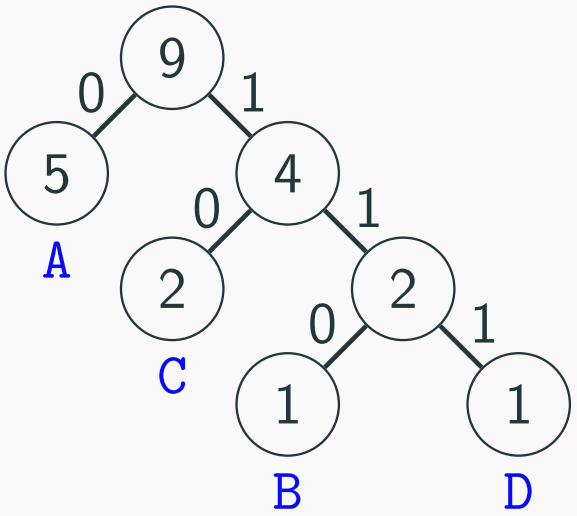
Complexity of lookup: O(length of word)

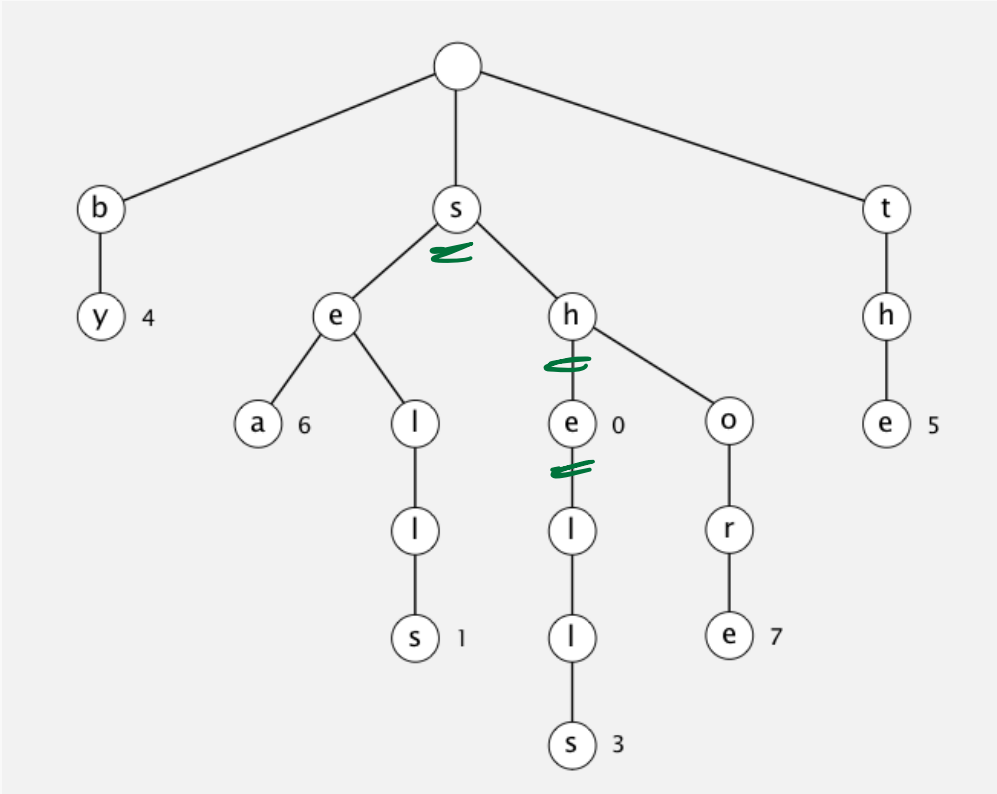
Lookup is independent of the number of strings in the trie!

Fast prefix search, but needs a lot of memory.

Can you tell the difference?

eg. she





Some questions

TQ

Find the longest word in a dictionary.

DFS
in trie data structure

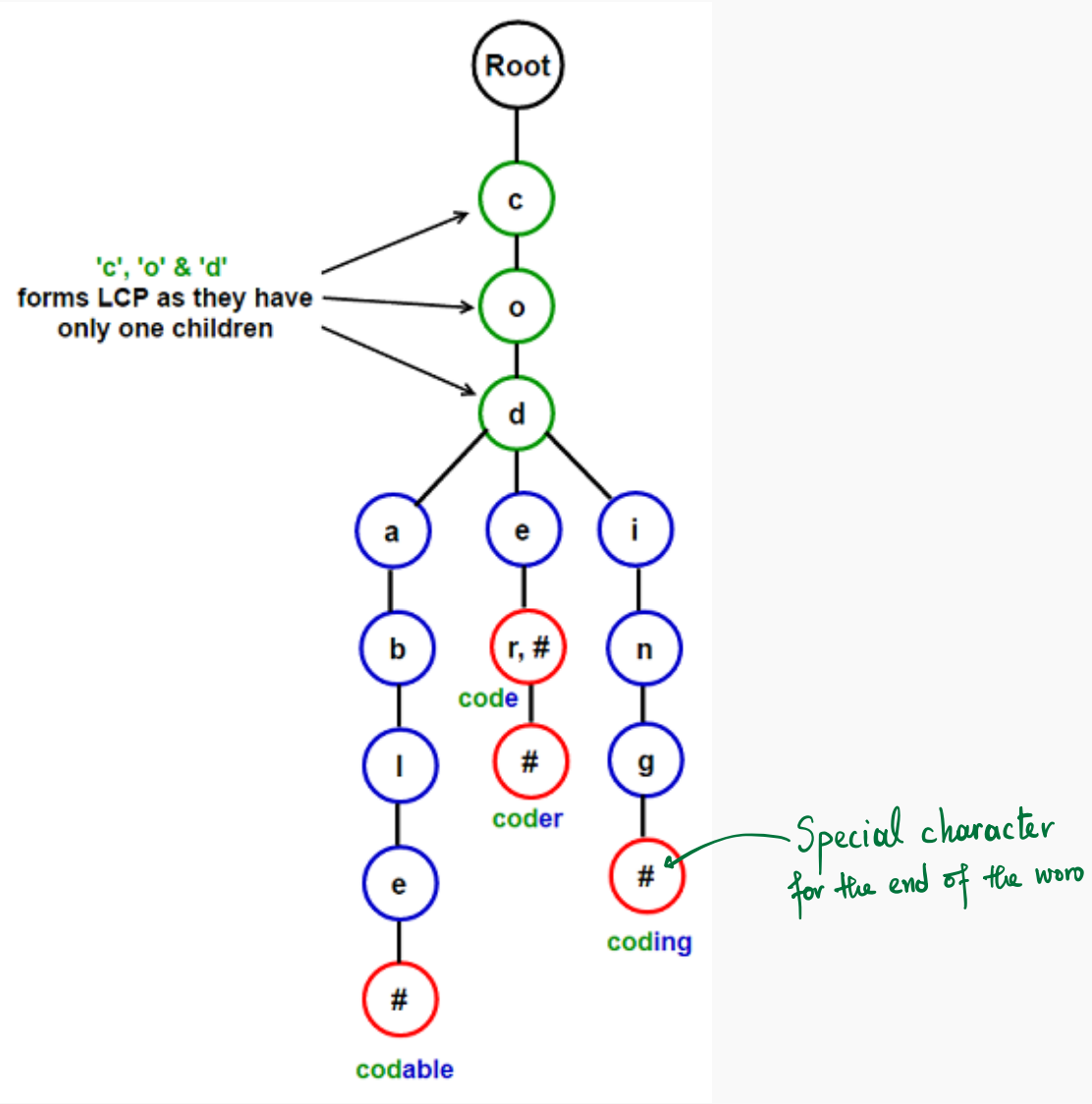
TQ

Write a web(?) application which provides instant search over the properties of some *some database*. The app does not wait for the user to press a submit button, but it dynamically updates the search results as input is typed.

TQ: Longest common prefix

LZW Compression

keys = [codable, code, coder, coding]



Applications of Trie

- dictionary lookup
- prefix searches
- auto complete feature in text editors
- command completion in IDE, editors, etc
- phone number/contacts searching
- IP lookups
- matching sentences (if a trie is based on a sequence of words, like Gmail autocompletion)
- finding positions of words in a text